## Question

expected: A = 59,100 km²  (M0 = 3.55e22 N·m)

The 2011 Tohoku earthquake had a moment magnitude Mw = 9.0. Using Mw = (2/3) log10(M0) - 10.7 (M0 in dyne·cm), calculate the seismic moment M0 in N·m. Then, with a shear modulus μ = 30 GPa and an average slip D = 20 m, estimate the rupture area A in km² using M0 = μ·A·D.

### Phase I · Pretrained base

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In order to estimate the fault dimensions, we can use empirical
relations proposed by Wells and Coppersmith [1994] for subduction
zones... The total slip deficit accumulates at a rate of 8 cm/yr
along the Japan Trench subduction boundary, leading to an estimated
recurrence interval of approximately 600 years for Mw 9.0 events. In
this study, we evaluate the spatial distribution of coseismic
slip...
```

✗ **continues the paper; no answer**

### Phase II · SFT

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<think>First compute seismic moment from magnitude: Mw = 9.0.
1.5*(9.0+10.7) = 29.55. So M0 = 10^29.55 dyne·cm = 3.55e22 N·m.
Rupture area A = M0 / (μ * D), where μ = 3e10 N/m², D = 20 m. So A =
3.55e22 / (3e10 * 20) = 5.92e10 m² = 59,200 km². Plugging in values:
A = 5.92e10 m² = 59,200 km². Plugging in values...
```

✗ **right idea, but the answer never finishes**

### Phase III · GRPO

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<think>M0 = 10^(1.5*(9.0+10.7)) dyne·cm. 1 dyne·cm = 1e-7 N·m, so M0
= 10^29.55 dyne·cm = 3.55e22 N·m. A = M0 / (μ * D) = 3.55e22 / (3e10
* 20) = 5.91e10 m² = 59,100 km².</think>
The seismic moment is about 3.55 × 10^22 N·m, and the rupture area
is about 59,100 km².
```

✓ **closes the reasoning and states the answer**

### Phase IV + vote@8 · Executed equations (deployed)

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<think>
Mw = 9.0 [-]
log_M0 = (3/2) * (Mw + 10.7) [-]
M0 = 10^log_M0 [dyne cm]
M0_Nm = M0 / 10^7 [N m]
mu = 3*10^10 [N/m^2]
D = 20 [m]
A_m2 = M0_Nm / (mu * D) [m^2]
A_km2 = A_m2 / 10^6 [km^2]
</think>Answer: 5.914e+04 km^2.
```

✓ **equations executed; 8 of 8 samples agree**